CLASS 10 — MATHS · FORMULA SHEET

Chapter 13: **Statistics** | Quick revision, one page

Class mark and basics

- Class mark $x_i = \dfrac{\text{upper limit} + \text{lower limit}}{2}$; class size h = upper limit − lower limit.
- $n = \sum f_i$ = total frequency.

Mean of grouped data

- Direct method: $\bar{x} = \dfrac{\sum f_i x_i}{\sum f_i}$.
- Assumed mean method: $\bar{x} = a + \dfrac{\sum f_i d_i}{\sum f_i}$, $d_i = x_i - a$.
- Step-deviation method: $\bar{x} = a + h\,\dfrac{\sum f_i u_i}{\sum f_i}$, $u_i = \dfrac{x_i - a}{h}$.
- Ungrouped data: mean $= \dfrac{\text{sum of observations}}{\text{number of observations}}$.

Mode of grouped data

- Modal class = class with the highest frequency.
- Mode $= l + \dfrac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h$.
- l = lower limit of modal class, f_1 = its frequency, f_0 = frequency of preceding class, f_2 = frequency of succeeding class, h = class size.

Median of grouped data

- Make a cumulative frequency (cf) column. Median class = class whose cf is just $\geq \dfrac{n}{2}$.
- Median $= l + \dfrac{\frac{n}{2} - cf}{f} \times h$.
- l = lower limit of median class, cf = cumulative frequency of the class preceding it, f = frequency of median class, h = class size.

Remember

- Classes must be continuous (exclusive); convert inclusive classes by adjusting limits by half the gap.
- Mean, median and mode are in the same units as the data.